

Trainings ▾

Install

with Mechanically Actuated Injector

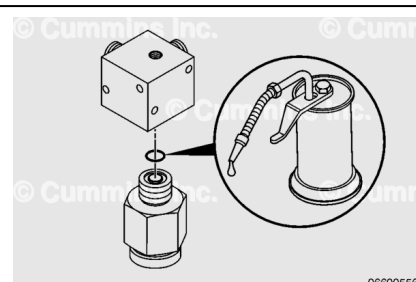
Note : This portion of the Install section is for engines with mechanically actuated injectors and the new style fuel drain line bracket.

Check that the o-ring has been fitted to the check valve.

Lubricate the o-ring with clean engine oil.

install the check valve into the tee line connector and tighten. It will be necessary to hold the line connector in a vise.

Torque Value: 60 n•m [44 ft-lb]



Install the o-ring to the coupling.

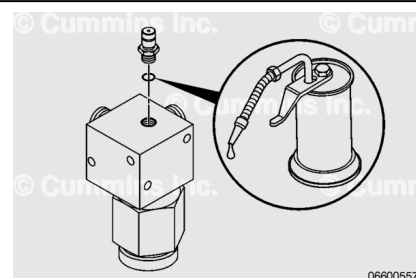
Lubricate the o-ring with clean engine oil.

Install the coupling in the same location and orientation as it was removed.

Torque Value:

Fuel Line Coupling

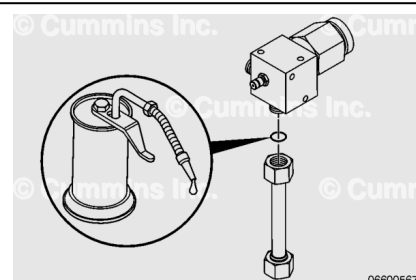
1.11 n•m [97 in-lb]



Install an o-ring to the tee line connector where the fuel drain line connects. Use clean engine oil to keep the o-rings in place.

Hold the drain line in the upright position to make sure the o-ring does **not** get dislodged. Install the fuel drain line to the tee line connector.

Hand tighten **only** at this time.

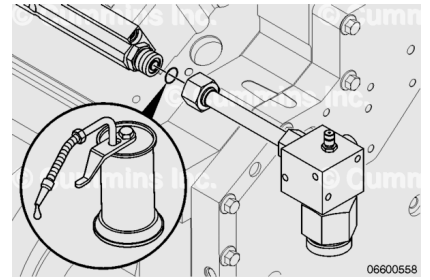


Install an o-ring onto the face of the male connector at the front of the right bank fuel manifold. Lubricate the o-ring with clean engine oil.

Install the tee line connector assembly to the male connector as shown in the illustration.

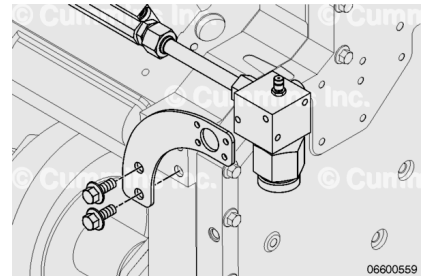
Hand tighten **only** at this time.

The check valve **must** be in the same location and orientation as it was removed.



Install the support bracket to the front mounting holes in the side of the front gear housing with two mounting capscrews.

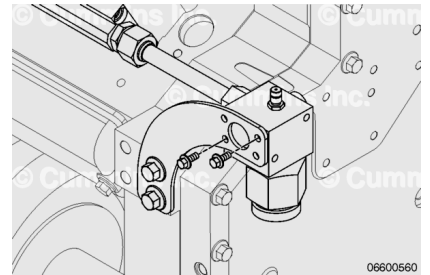
Hand tighten **only** at this time.



Install the support bracket to the tee line connector with two mounting capscrews.

Hand tighten **only** at this time.

Tighten the fuel drain line nut nearest the fuel manifold, while holding the male connector in place.



Torque Value:

Fuel Drain Line Nut Nearest Fuel Manifold

1. 65 n•m [48 ft-lb]

Tighten the fuel drain line nut nearest the connector block.

Torque Value:

Fuel Drain Line Nut Nearest Connector Block

1. 65 n•m [48 ft-lb]

Tighten the two capscrews that secure the tee line connector to the bracket.

Torque Value:

Bracket to Connector Capscrews

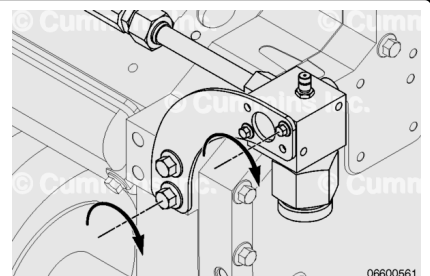
1. 18 n•m [159 in-lb]

Tighten the two capscrews that secure the bracket to the front of the gear housing.

Torque Value:

Bracket to Gear Housing Capscrews

1. 195 n•m [144 ft-lb]



Check the fuel drain line and nuts at each end of the line.

Install an o-ring onto the male connector at the tee line and an o-ring to the male connector at the fuel manifold.

Lubricate the o-rings with clean engine oil.

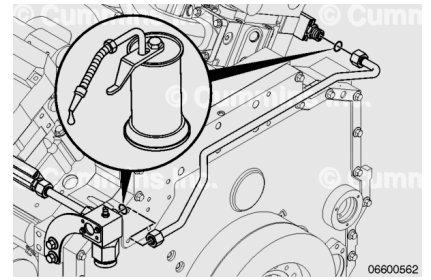
Install the fuel drain line to the male connector at the tee line. Make sure the o-ring does **not** get dislodged. Tighten the fuel drain line nut at the tee line.

Torque Value: 120 n•m [89 ft-lb]

Install the fuel drain line nut at the fuel manifold male connector and tighten.

Torque Value: 120 n•m [89 ft-lb]

Install the fuel return line. Refer to the OEM service manual.

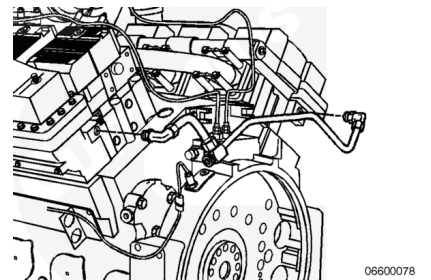


Note : This portion of the Install section is for engines with mechanically actuated injectors and the old style fuel drain line bracket.

Note : The option illustrated is a left rear fuel drain location. The fuel drain can also be located at the right rear, left front, and right front.

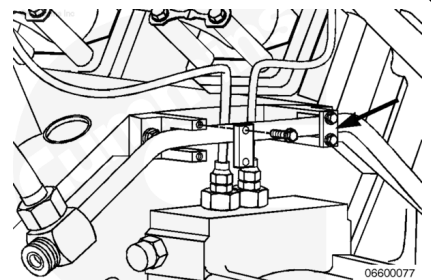
Install and tighten the fuel drain line to the fuel manifold.

Torque Value: 45 n•m [33 ft-lb]



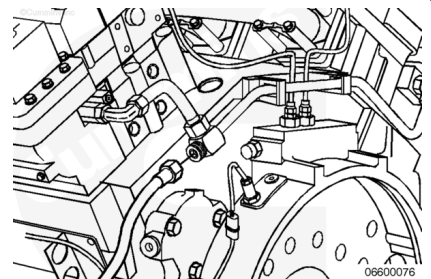
Install the support block clamp over the fuel drain line and tighten.

Torque Value: 9 n•m [80 in-lb]



Install and tighten the drain hose to the fuel drain line.

Torque Value: 45 n•m [33 ft-lb]

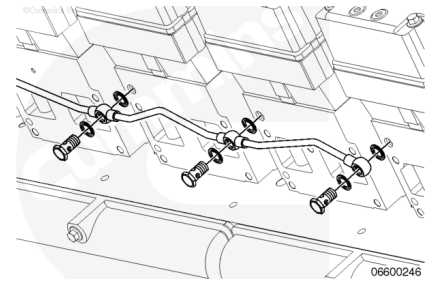


with Electronically Actuated Injector

Install the fuel drain manifold, eight banjo capscrews, and new sealing washers.

Torque Value:

Fuel Drain Manifold Capscrews
1. 30 n•m [22 ft-lb]

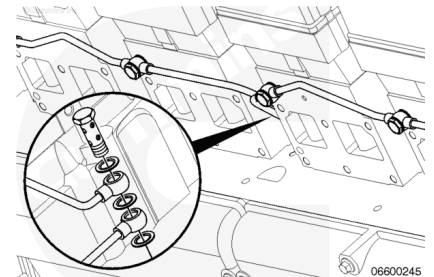


Install the double banjo capscrew and new sealing washers that secure the fuel drain manifold to the block.

Torque Value:

Fuel Drain Manifold
1. 40 n•m [30 ft-lb]

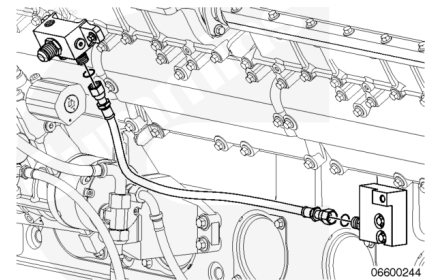
Repeat these procedures on the opposite cylinder bank.



Install the fuel drain line that connects the drain manifold block to the fuel block manifold. Use new o-rings at each end.

Torque Value:

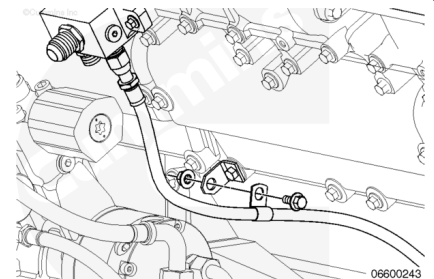
Fuel Drain Line
1. 45 n•m [33 ft-lb]



Install the screw, nut and p-clips which clamp the fuel drain line between the drain manifold block and the fuel block manifold, to the brace that secures it to the cylinder block.

Torque Value:

P-Clip Capscrew
1. 22 n•m [195 in-lb]



Install the banjo capscrew and new sealing washers securing the air bleed line from the drain manifold block (2) to the fuel pump (1).

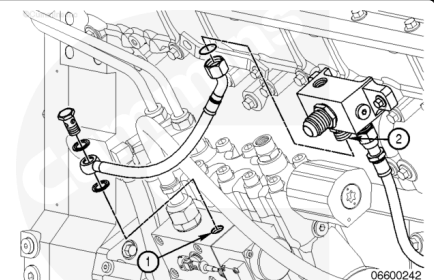
Torque Value:

Air Bleed Line
1. 30 n•m [22 ft-lb]

Install the air bleed line to the drain manifold block. Use a new o-ring.

Torque Value:

Air Bleed Line



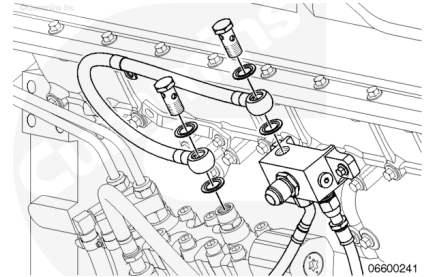
1. 45 n•m [33 ft-lb]

Install the banjo capscrews and new sealing washers securing the drain line between the mechanical dump valve and the drain manifold block.

Torque Value:

Fuel Drain Line Banjo Capscrew

1. 48 n•m [35 ft-lb]

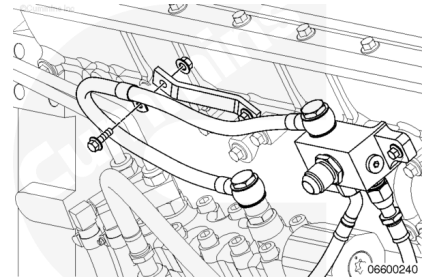


Install the capscrew and p-clip which clamp the drain line between the mechanical dump valve and the drain manifold block, to the brace that secures it to the cylinder block.

Torque Value:

P-Clip Capscrew

1. 22 n•m [195 in-lb]



For Generator Model DQKAN with Low NOx Emissions Capability

The procedure for installation of the main fuel manifold, if necessary, is the same as detailed above for standard product.

Install new o-rings.

Connect the fitting at the drain manifold block and fuel manifold block.

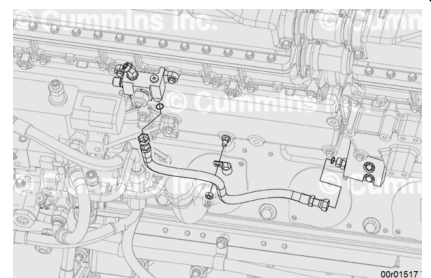
Install p-clip to the camshaft follower housing.

Torque Value:

P-Clip 43 n•m [32 ft-lb]

Torque Value:

Hose Fittings 55 n•m [41 ft-lb]

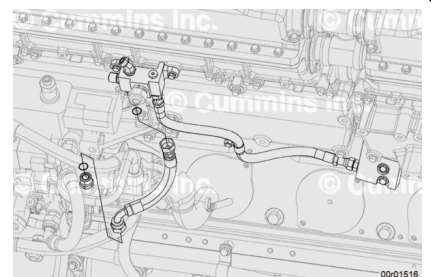


Install new o-rings.

Install the fitting at the fuel pump gerotor and drain manifold block.

Torque Value:

Hose Fittings 60 n•m [44 ft-lb]



Install new o-rings.

Install the line, securing the fittings at the high-pressure pump and fuel drain manifold hand tight.

Install the line to the block clamp securing the Pump return line to the bracket and tighten.

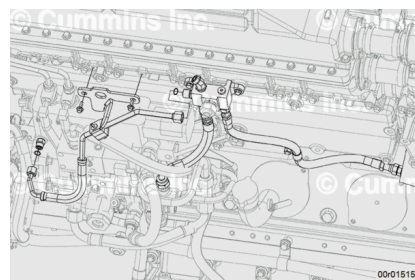
Torque the fittings securing the line at the high-pressure pump and fuel drain manifold.

Torque Value:

Hose Fittings 55 n•m [41 ft-lb]

Torque Value:

Hose Bracket 10 n•m [89 in-lb]



Install new o-rings.

Connect and torque the fittings at the fuel pressure relief valve and the fuel drain manifold.

Torque Value:

Hose Fittings 45 n•m [33 ft-lb]

